

(%i12) **load(vect)\$; /* Vector package */**

/* Symbols and functions */

declare(λ , constant)\$;

declare(g , constant)\$;

declare(Ω , constant)\$;

declare(β , constant)\$;

declare(m , constant)\$;

assume($t \geq 0$, $\lambda > 0$, $g > 0$, $\Omega > 0$, $\beta > 0$, $m > 0$)\$;

depends([x , y , z], t)\$;

rv: [$x(t)$, $y(t)$, $z(t)$]\$; /* Vector position */

gv: [0 , 0 , $-g$]\$; /* Gravity */

omegav: [0 , $\Omega \cdot \cos(\lambda)$, $\Omega \cdot \sin(\lambda)$]\$; /* Angular velocity vector of Earth */

eq: diff(rv, t, 2) = gv - 2 * express(omegav ~ diff(rv, t, 1)) - express(omegav ~ (express(omegav ~

eq
$$\begin{aligned} \left[\frac{d^2}{dt^2} x(t), \frac{d^2}{dt^2} y(t), \frac{d^2}{dt^2} z(t) \right] = & \left[- \left(2 \left(\Omega \cos(\lambda) \left(\frac{d}{dt} z(t) \right) - \Omega \sin(\lambda) \left(\frac{d}{dt} y(t) \right) \right) \right) - \right. \\ & \frac{\beta \left(\frac{d}{dt} x(t) \right)}{m} + \Omega^2 \sin^2(\lambda) x(t) + \Omega^2 \cos^2(\lambda) x(t), - \left(\frac{\beta \left(\frac{d}{dt} y(t) \right)}{m} \right) - 2 \Omega \sin(\lambda) \left(\frac{d}{dt} x(t) \right) - \\ & \Omega \sin(\lambda) (\Omega \cos(\lambda) z(t) - \Omega \sin(\lambda) y(t)), - \left(\frac{\beta \left(\frac{d}{dt} z(t) \right)}{m} \right) + 2 \Omega \cos(\lambda) \left(\frac{d}{dt} x(t) \right) + \Omega \cos(\lambda) \\ & \left. (\Omega \cos(\lambda) z(t) - \Omega \sin(\lambda) y(t)) - g \right] \end{aligned}$$

(%i13) **eq: subst([$\Omega^2 = 0$], expand(eq)); /* Linearizing the system */**

eq
$$\begin{aligned} \left[\frac{d^2}{dt^2} x(t), \frac{d^2}{dt^2} y(t), \frac{d^2}{dt^2} z(t) \right] = & \left[- \left(2 \Omega \cos(\lambda) \left(\frac{d}{dt} z(t) \right) \right) + 2 \Omega \sin(\lambda) \left(\frac{d}{dt} y(t) \right) - \right. \\ & \frac{\beta \left(\frac{d}{dt} x(t) \right)}{m}, - \left(\frac{\beta \left(\frac{d}{dt} y(t) \right)}{m} \right) - 2 \Omega \sin(\lambda) \left(\frac{d}{dt} x(t) \right), - \left(\frac{\beta \left(\frac{d}{dt} z(t) \right)}{m} \right) + 2 \Omega \cos(\lambda) \\ & \left. \left(\frac{d}{dt} x(t) \right) - g \right] \end{aligned}$$

(%i16) **/* Differential equations */**

eqx: lhs(eq)[1] = rhs(eq)[1];

eqy: lhs(eq)[2] = rhs(eq)[2];

eqz: lhs(eq)[3] = rhs(eq)[3];

eqx
$$\frac{d^2}{dt^2} x(t) = - \left(2 \Omega \cos(\lambda) \left(\frac{d}{dt} z(t) \right) \right) + 2 \Omega \sin(\lambda) \left(\frac{d}{dt} y(t) \right) - \frac{\beta \left(\frac{d}{dt} x(t) \right)}{m}$$

$$\text{eqy} \quad \frac{d^2}{dt^2} y(t) = - \left(\frac{\beta \left(\frac{d}{dt} y(t) \right)}{m} \right) - 2 \Omega \sin(\lambda) \left(\frac{d}{dt} x(t) \right)$$

$$\text{eqz} \quad \frac{d^2}{dt^2} z(t) = - \left(\frac{\beta \left(\frac{d}{dt} z(t) \right)}{m} \right) + 2 \Omega \cos(\lambda) \left(\frac{d}{dt} x(t) \right) - g$$

(%i22) /* Introducing Laplace Transform */

assume(s, complex)\$;

depends([X, Y, Z], s)\$;

subs: [

laplace(x(t), t, s) = X(s),

laplace(y(t), t, s) = Y(s),

laplace(z(t), t, s) = Z(s),

at(x(t), [t = 0]) = x_0,

at(y(t), [t = 0]) = y_0,

at(z(t), [t = 0]) = z_0,

at('diff(x(t), t, 1), [t = 0]) = a_0,

at('diff(y(t), t, 1), [t = 0]) = b_0,

at('diff(z(t), t, 1), [t = 0]) = c_0

]\$;

laplace_eqx: subst(subs, laplace(eqy, t, s));

laplace_eqy: subst(subs, laplace(eqy, t, s));

laplace_eqz: subst(subs, laplace(eqz, t, s));

$$\text{laplace_eqx} - (s x_0) + s^2 X(s) - a_0 = - (2 \Omega \cos(\lambda) (s Z(s) - z_0)) + 2 \Omega \sin(\lambda) (s Y(s) - y_0) - \frac{\beta (s X(s) - x_0)}{m}$$

$$\text{laplace_eqy} - (s y_0) + s^2 Y(s) - b_0 = - \left(\frac{\beta (s Y(s) - y_0)}{m} \right) - 2 \Omega \sin(\lambda) (s X(s) - x_0)$$

$$\text{laplace_eqz} - (s z_0) + s^2 Z(s) - c_0 = - \left(\frac{\beta (s Z(s) - z_0)}{m} \right) + 2 \Omega \cos(\lambda) (s X(s) - x_0) - \frac{g}{s}$$

(%i23) **laplace_solution: solve([laplace_eqx, laplace_eqy, laplace_eqz], [X(s), Y(s), Z(s)]); /* Find X(s), Y(s), Z(s) */**

$$\text{laplace_solution} [[X(s) = \left(m^2 s^3 + 2 m \beta s^2 + \left(4 m^2 \Omega^2 \sin(\lambda)^2 + 4 m^2 \Omega^2 \cos(\lambda)^2 + \beta^2 \right) s \right) x_0 + m^2 a_0 s^2 + \left(- \left(2 m^2 \Omega \cos(\lambda) c_0 \right) + 2 m^2 \Omega \sin(\lambda) b_0 + m \beta a_0 \right) s + 2 g m^2 \Omega \cos(\lambda) /$$

$$, Y(s) = (m^3 s^4 + 3 m^2 \beta s^3 +$$

$$\left(4 m^3 \Omega^2 \sin(\lambda)^2 + 4 m^3 \Omega^2 \cos(\lambda)^2 + 3 m \beta^2 \right) s^2 +$$

$$\left(4 m^2 \Omega^2 \beta \sin(\lambda)^2 + 4 m^2 \Omega^2 \beta \cos(\lambda)^2 + \beta^3 \right) s) y_0 + m^3 b_0 s^3 +$$

$$\left(2 m^2 \beta b_0 - 2 m^3 \Omega \sin(\lambda) a_0 \right) s^2 +$$

$$\begin{aligned} & \left(4 m^3 \Omega^2 \cos(\lambda) \sin(\lambda) c_0 + 4 m^3 \Omega^2 \cos(\lambda)^2 b_0 + m \beta^2 b_0 - 2 m^2 \Omega \beta \sin(\lambda) a_0 \right) s - 4 g m^3 \Omega^2 \\ & \cos(\lambda) \sin(\lambda) / m^3 s^5 + 3 m^2 \beta s^4 + \left(4 m^3 \Omega^2 \sin(\lambda)^2 + 4 m^3 \Omega^2 \cos(\lambda)^2 + 3 m \beta^2 \right) s^3 + \\ & \left(4 m^2 \Omega^2 \beta \sin(\lambda)^2 + 4 m^2 \Omega^2 \beta \cos(\lambda)^2 + \beta^3 \right) s^2, Z(s) = (m^3 s^4 + 3 m^2 \beta s^3 + \\ & \left(4 m^3 \Omega^2 \sin(\lambda)^2 + 4 m^3 \Omega^2 \cos(\lambda)^2 + 3 m \beta^2 \right) s^2 + \\ & \left(4 m^2 \Omega^2 \beta \sin(\lambda)^2 + 4 m^2 \Omega^2 \beta \cos(\lambda)^2 + \beta^3 \right) s) z_0 + m^3 c_0 s^3 + \\ & \left(2 m^2 \beta c_0 + 2 m^3 \Omega \cos(\lambda) a_0 - g m^3 \right) s^2 + \\ & \left(4 m^3 \Omega^2 \sin(\lambda)^2 c_0 + m \beta^2 c_0 + 4 m^3 \Omega^2 \cos(\lambda) \sin(\lambda) b_0 + 2 m^2 \Omega \beta \cos(\lambda) a_0 - 2 g m^2 \beta \right) s - \\ & 4 g m^3 \Omega^2 \sin(\lambda)^2 - g m \beta^2 / m^3 s^5 + 3 m^2 \beta s^4 + \\ & \left(4 m^3 \Omega^2 \sin(\lambda)^2 + 4 m^3 \Omega^2 \cos(\lambda)^2 + 3 m \beta^2 \right) s^3 + \\ & \left(4 m^2 \Omega^2 \beta \sin(\lambda)^2 + 4 m^2 \Omega^2 \beta \cos(\lambda)^2 + \beta^3 \right) s^2] \end{aligned}$$

(%i27) /* Simplifications */

laplace_solution: subst([$\Omega^2 = 0$], expand(laplace_solution));

X_laplace: factor(trigsimp(rhs(laplace_solution[1][1])));

Y_laplace: factor(trigsimp(rhs(laplace_solution[1][2])));

Z_laplace: factor(trigsimp(rhs(laplace_solution[1][3])));

X_laplace
$$\frac{m^2 s^3 x_0 + 2 m \beta s^2 x_0 + \beta^2 s x_0 + m^2 a_0 s^2 - 2 m^2 \Omega \cos(\lambda) c_0 s + 2 m^2 \Omega \sin(\lambda) b_0 s + m \beta a_0 s + 2 g m^2 \Omega \cos(\lambda)}{s^2 (m s + \beta)}$$

Y_laplace
$$\frac{m^2 s^2 y_0 + 2 m \beta s y_0 + \beta^2 y_0 + m^2 b_0 s + m \beta b_0 - 2 m^2 \Omega \sin(\lambda) a_0}{s^2 (m s + \beta)}$$

Z_laplace
$$\frac{m^2 s^3 z_0 + 2 m \beta s^2 z_0 + \beta^2 s z_0 + m^2 c_0 s^2 + m \beta c_0 s + 2 m^2 \Omega \cos(\lambda) a_0 s - g m^2 s - g m \beta}{s^2 (m s + \beta)}$$

(%i28) **x:** ratsimp(ilt((X_laplace), s, t)); /* x(t) */

x
$$\begin{aligned} & e^{-\left(\frac{\beta t}{m}\right)} \left(\frac{\beta t}{m} \beta^3 x_0 + \left(2 m \Omega \beta^2 \cos(\lambda) c_0 - 2 m \Omega \beta^2 \sin(\lambda) b_0 + \left(2 e^{\frac{\beta t}{m}} + 2 \right) g m^2 \Omega \beta \cos(\lambda) \right) t + \left(2 - 2 e^{\frac{\beta t}{m}} \right) \right. \\ & m^2 \Omega \beta \cos(\lambda) c_0 + \left(2 e^{\frac{\beta t}{m}} - 2 \right) m^2 \Omega \beta \sin(\lambda) b_0 + \left(e^{\frac{\beta t}{m}} - 1 \right) m \beta^2 a_0 + \left(4 - 4 e^{\frac{\beta t}{m}} \right) g \\ & \left. m^3 \Omega \cos(\lambda) \right) / \beta^3 \end{aligned}$$

(%i29) **y:** ratsimp(ilt((Y_laplace), s, t)); /* y(t) */

y
$$e^{-\left(\frac{\beta t}{m}\right)}$$

$$\left(\%e^{\frac{\beta t}{m}} \beta^2 y_0 + 2 m \Omega \beta \sin(\lambda) a_0 t + \left(\%e^{\frac{\beta t}{m}} - 1 \right) m \beta b_0 + \left(2 - 2 \%e^{\frac{\beta t}{m}} \right) m^2 \Omega \sin(\lambda) a_0 \right) / \beta^2$$

→

(%i30) **z: ratsimp(ilt((Z_laplace), s, t)); /* z(t) */**

z

$$\%e^{-\left(\frac{\beta t}{m}\right)} \left(\%e^{\frac{\beta t}{m}} \beta^2 z_0 + \left(- (2 m \Omega \beta \cos(\lambda) a_0) - \%e^{\frac{\beta t}{m}} g m \beta \right) t + \left(\%e^{\frac{\beta t}{m}} - 1 \right) m \beta c_0 + \left(2 \%e^{\frac{\beta t}{m}} - 2 \right) m^2 \Omega \cos(\lambda) a_0 + \left(\%e^{\frac{\beta t}{m}} - 1 \right) g m^2 \right) / \beta^2$$

(%i37) **/* Setting initial conditions of the projectile */**

assume(0 <= φ , φ <= %pi)\$;

assume(0 <= θ , θ <= 2 · %pi)\$;

assume(v_0 > 0)\$;

subs: [/* Spherical coordinates */

x_0 = 0,

y_0 = 0,

z_0 = 0,

a_0 = v_0 · cos(θ) · sin(φ),

b_0 = v_0 · sin(θ) · sin(φ),

c_0 = v_0 · cos(φ)

];

/* General solutions */

x: ratsimp(subst(subs, x));

y: ratsimp(subst(subs, y));

z: ratsimp(subst(subs, z));

x

$$\%e^{-\left(\frac{\beta t}{m}\right)} \left(\left(2 m \Omega \beta^2 \sin(\lambda) t + \left(2 - 2 \%e^{\frac{\beta t}{m}} \right) m^2 \Omega \beta \sin(\lambda) \right) v_0 \sin(\theta) + \left(1 - \%e^{\frac{\beta t}{m}} \right) m \beta^2 v_0 \cos(\theta) \right) \sin(\varphi) + \left(\left(2 \%e^{\frac{\beta t}{m}} - 2 \right) m^2 \Omega \beta \cos(\lambda) - 2 m \Omega \beta^2 \cos(\lambda) t \right) v_0 \cos(\varphi) + \left(- \left(2 \%e^{\frac{\beta t}{m}} \right) - 2 \right) g m^2 \Omega \beta \cos(\lambda) t + \left(4 \%e^{\frac{\beta t}{m}} - 4 \right) g m^3 \Omega \cos(\lambda) \right) / \beta^3$$

y

$$\%e^{-\left(\frac{\beta t}{m}\right)} \left(\left(\%e^{\frac{\beta t}{m}} - 1 \right) m \beta v_0 \sin(\theta) + \left(2 m \Omega \beta \sin(\lambda) t + \left(2 - 2 \%e^{\frac{\beta t}{m}} \right) m^2 \Omega \sin(\lambda) \right) v_0 \cos(\theta) \right) \sin(\varphi) / \beta^2$$

$$z = -\left(\frac{\beta t}{m}\right) \left(\left(2 m \Omega \beta \cos(\lambda) t + \left(2 - 2 e^{-\frac{\beta t}{m}} \right) m^2 \Omega \cos(\lambda) \right) v_0 \cos(\theta) \sin(\varphi) + \left(1 - e^{-\frac{\beta t}{m}} \right) m \beta v_0 \cos(\varphi) + e^{-\frac{\beta t}{m}} g m \beta t + \left(1 - e^{-\frac{\beta t}{m}} \right) g m^2 / \beta^2 \right)$$

(%i40) /* East launch */

ratsimp(subst([θ = 0], x));

ratsimp(subst([θ = 0], y));

ratsimp(subst([θ = 0], z));

$$\begin{aligned} & e^{-\left(\frac{\beta t}{m}\right)} \left(\left(e^{-\frac{\beta t}{m}} - 1 \right) m \beta^2 v_0 \sin(\varphi) + \left(2 m \Omega \beta^2 \cos(\lambda) t + \left(2 - 2 e^{-\frac{\beta t}{m}} \right) m^2 \Omega \beta \cos(\lambda) \right) v_0 \cos(\varphi) + \left(2 e^{-\frac{\beta t}{m}} + 2 \right) g m^2 \Omega \beta \cos(\lambda) t + \left(4 - 4 e^{-\frac{\beta t}{m}} \right) g m^3 \Omega \cos(\lambda) \right) / \beta^3 \\ & \frac{e^{-\left(\frac{\beta t}{m}\right)} \left(2 m \Omega \beta \sin(\lambda) t + \left(2 - 2 e^{-\frac{\beta t}{m}} \right) m^2 \Omega \sin(\lambda) \right) v_0 \sin(\varphi)}{\beta^2} \end{aligned}$$

$$\begin{aligned} & -\left(\frac{\beta t}{m}\right) \left(\left(2 m \Omega \beta \cos(\lambda) t + \left(2 - 2 e^{-\frac{\beta t}{m}} \right) m^2 \Omega \cos(\lambda) \right) v_0 \sin(\varphi) + \left(1 - e^{-\frac{\beta t}{m}} \right) m \beta v_0 \cos(\varphi) + e^{-\frac{\beta t}{m}} g m \beta t + \left(1 - e^{-\frac{\beta t}{m}} \right) g m^2 / \beta^2 \right) \end{aligned}$$

(%i43) /* North launch */

ratsimp(subst([θ = %pi / 2], x));

ratsimp(subst([θ = %pi / 2], y));

ratsimp(subst([θ = %pi / 2], z));

$$\begin{aligned} & -\left(\frac{\beta t}{m}\right) \left(\left(2 m \Omega \beta^2 \sin(\lambda) t + \left(2 - 2 e^{-\frac{\beta t}{m}} \right) m^2 \Omega \beta \sin(\lambda) \right) v_0 \sin(\varphi) + \left(\left(e^{-\frac{\beta t}{m}} - 2 \right) m^2 \Omega \beta \cos(\lambda) - 2 m \Omega \beta^2 \cos(\lambda) t \right) v_0 \cos(\varphi) + \left(-\left(e^{-\frac{\beta t}{m}} \right) - 2 \right) g m^2 \Omega \beta \cos(\lambda) t + \left(4 e^{-\frac{\beta t}{m}} - 4 \right) g m^3 \Omega \cos(\lambda) \right) / \beta^3 \\ & \frac{e^{-\left(\frac{\beta t}{m}\right)} \left(e^{-\frac{\beta t}{m}} - 1 \right) m v_0 \sin(\varphi)}{\beta} \end{aligned}$$

(%o43)
$$\frac{-\left(\frac{\beta t}{m}\right) \left(\left(\frac{\beta t}{m} - 1 \right) m \beta v_0 \cos(\varphi) - \frac{\beta t}{m} g m \beta t + \left(\frac{\beta t}{m} - 1 \right) g m^2 \right)}{\beta^2}$$

(%i46) /* West launch */

ratsimp(subst([θ = %pi], x));

ratsimp(subst([θ = %pi], y));

ratsimp(subst([θ = %pi], z));

(%o44)
$$-\left(\frac{\beta t}{m}\right) \left(\left(\frac{\beta t}{m} - 1 \right) m \beta^2 v_0 \sin(\varphi) + \left(\left(\frac{\beta t}{m} - 2 \right) m^2 \Omega \beta \cos(\lambda) - 2 m \Omega \beta^2 \cos(\lambda) t \right) v_0 \cos(\varphi) + \left(- \left(\frac{\beta t}{m} - 2 \right) g m^2 \Omega \beta \cos(\lambda) t + \left(\frac{\beta t}{m} - 4 \right) g m^3 \Omega \cos(\lambda) \right) / \beta^3 \right)$$

(%o45)
$$-\left(\frac{-\left(\frac{\beta t}{m}\right) \left(2 m \Omega \beta \sin(\lambda) t + \left(\frac{\beta t}{m} - 2 \right) m^2 \Omega \sin(\lambda) \right) v_0 \sin(\varphi)}{\beta^2} \right)$$

(%o46)
$$\frac{-\left(\frac{\beta t}{m}\right) \left(\left(2 m \Omega \beta \cos(\lambda) t + \left(\frac{\beta t}{m} - 2 \right) m^2 \Omega \cos(\lambda) \right) v_0 \sin(\varphi) + \left(\frac{\beta t}{m} - 1 \right) m \beta v_0 \cos(\varphi) - \frac{\beta t}{m} g m \beta t + \left(\frac{\beta t}{m} - 1 \right) g m^2 \right) / \beta^2}$$

(%i49) /* South launch */

ratsimp(subst([θ = - %pi / 2], x));

ratsimp(subst([θ = - %pi / 2], y));

ratsimp(subst([θ = - %pi / 2], z));

(%o47)
$$\frac{-\left(\frac{\beta t}{m}\right) \left(\left(2 m \Omega \beta^2 \sin(\lambda) t + \left(\frac{\beta t}{m} - 2 \right) m^2 \Omega \beta \sin(\lambda) \right) v_0 \sin(\varphi) + \left(2 m \Omega \beta^2 \cos(\lambda) t + \left(\frac{\beta t}{m} - 2 \right) m^2 \Omega \beta \cos(\lambda) \right) v_0 \cos(\varphi) + \left(\frac{\beta t}{m} + 2 \right) g m^2 \Omega \beta \cos(\lambda) t + \left(\frac{\beta t}{m} - 4 \right) g m^3 \Omega \cos(\lambda) \right) / \beta^3}$$

(%o48)
$$-\left(\frac{-\left(\frac{\beta t}{m}\right) \left(\frac{\beta t}{m} - 1 \right) m v_0 \sin(\varphi)}{\beta} \right)$$

(%o49)
$$\frac{{\%e}^{-\left(\frac{\beta t}{m}\right)}\left(\left({\%e}^{\frac{\beta t}{m}}-1\right)m\beta v_0\cos(\varphi)-{\%e}^{\frac{\beta t}{m}}gm\beta t+\left({\%e}^{\frac{\beta t}{m}}-1\right)gm^2\right)}{\beta^2}$$

(%i52) /* Pure vertical launch */
ratsimp(subst([$\varphi = 0$], x));
subst([$\varphi = 0$], y);
ratsimp(subst([$\varphi = 0$], z));

(%o50)
$${\%e}^{-\left(\frac{\beta t}{m}\right)}\left(2m\Omega\beta^2\cos(\lambda)t+\left(2-2{\%e}^{\frac{\beta t}{m}}\right)m^2\Omega\beta\cos(\lambda)\right)v_0+\left(2{\%e}^{\frac{\beta t}{m}}+2\right)gm^2\Omega\beta\cos(\lambda)t+\left(4-4{\%e}^{\frac{\beta t}{m}}\right)gm^3\Omega\cos(\lambda))/\beta^3$$

(%o51) 0

(%o52)
$$\frac{{\%e}^{-\left(\frac{\beta t}{m}\right)}\left(\left({\%e}^{\frac{\beta t}{m}}-1\right)m\beta v_0-{\%e}^{\frac{\beta t}{m}}gm\beta t+\left({\%e}^{\frac{\beta t}{m}}-1\right)gm^2\right)}{\beta^2}$$

(%i53) **T: factor(solve(z = 0, t))[1];** /* Computing the exact time of flight isn't straightforward. Let's do it n

T
$$t = \frac{\left({\%e}^{\frac{\beta t}{m}}-1\right)(2m\Omega\cos(\lambda)v_0\cos(\theta)\sin(\varphi)+\beta v_0\cos(\varphi)+gm)}{\beta\left(2\Omega\cos(\lambda)v_0\cos(\theta)\sin(\varphi)+{\%e}^{\frac{\beta t}{m}}g\right)}$$

(%i56) /* Going numerically*/

T: rhs(T) - t = 0\$;

nums: [

g = 9.81, /* m / s^2*/

$\Omega = 7.29 \cdot 10^{(-5)}$, /* rad / s*/

$\lambda = \%pi / 4.0$, /* rad*/

v_0 = 100.0, /* m / s*/

$\theta = 0.0$, /* rad*/

$\varphi = \%pi / 4.0$, /* rad*/

m = 1.0, /* Kg*/

$\beta = 0.05$ /* Kg / s */

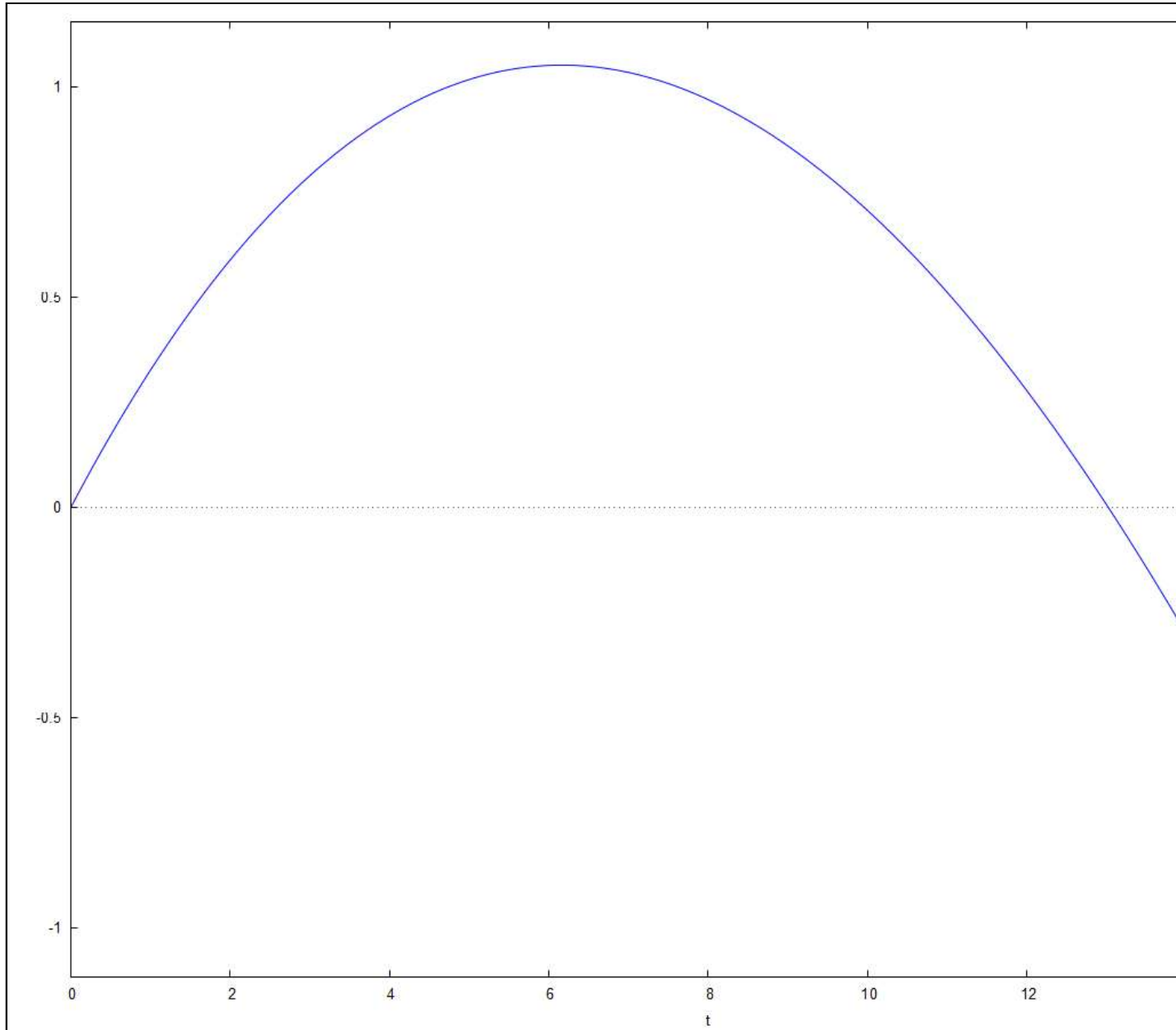
];

T: subst(nums, T);

T
$$\frac{20.0({\%e}^{0.05t}-1)(0.014580000000000003\cos(0.25\pi)\sin(0.25\pi)+5.0\cos(0.25\pi)+9.81)}{0.014580000000000003\cos(0.25\pi)\sin(0.25\pi)+9.81{\%e}^{0.05t}} - t = 0$$

(%i57) **wxplot2d([lhs(T)], [t, 0, 16]);** /* Plotting to isolate the root */

(%t57)



(%o57)

(%i58) **T: find_root(T, t, 12.0, 14.0, abserr = 1e-06, relerr = 1e-03);** /* Time of flight computed [s] */

T 13.02154687343301

(%i60) /* x (east) and y (north) final positions [m]*/

subst([t = T], subst(nums, x)), numer;

subst([t = T], subst(nums, y)), numer;

(%o59) 676.5906282896381

(%o60) -0.4052952744781692

(%i61) **max_t: solve(factor(diff(z, t, 1)) = 0, t)[1]**/* Maximum elevation. Another root finding problem */;

$$\text{max_t} = \frac{-(\beta v_0 \cos(\varphi)) + e^{\frac{\beta t}{m}} g m - g m}{2 \Omega \beta \cos(\lambda) v_0 \cos(\theta) \sin(\varphi)}$$

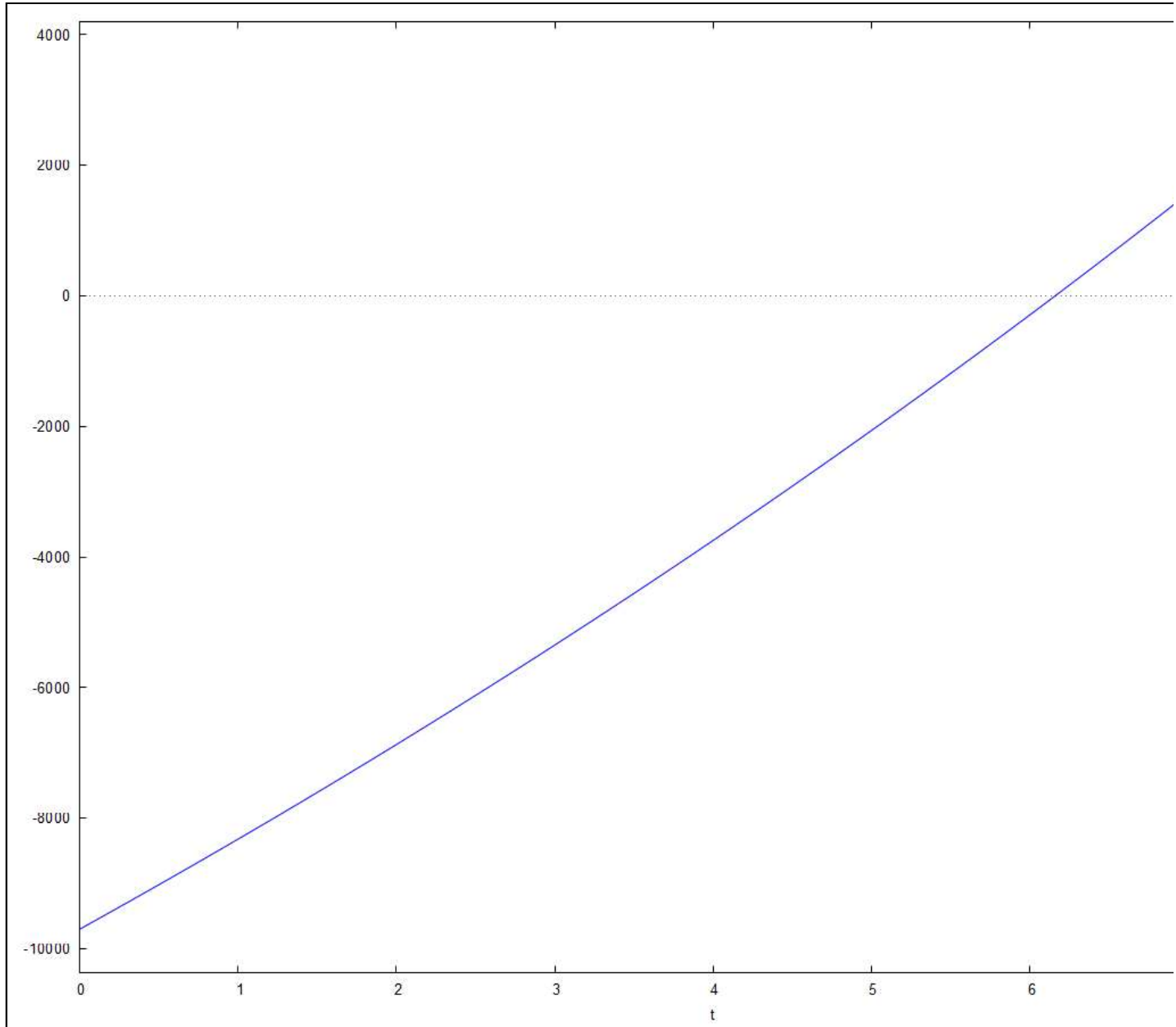
(%i63) **max_t: rhs(max_t) - t = 0\$;**

max_t: subst(nums, max_t);

$$\text{max_t} = \frac{1371.742112482853 \left(-(5.0 \cos(0.25 \pi)) + 9.81 e^{0.05 t} - 9.81 \right)}{\cos(0.25 \pi) \sin(0.25 \pi)} - t = 0$$

(%i64) **wxplot2d([lhs(max_t)], [t, 0, 8]);** /* Plotting to isolate the root */

(%t64)



```
(%o64)
(%i65) max_t: find_root(max_t, t, 6.0, 7.0, abserr = 1e-06, relerr = 1e-03); /* Time of maximum elevation co
max_t 6.158954199314835
(%i66) subst([t = max_t], subst(nums, z)), numer; /* Maximum elevation [m] */
(%o66) 206.59958884906786
(%i67) wxdraw3d( /* Full trajectory */
    color = red,
    line_width = 2,
    view = [60, 30],
    parametric(subst(nums, x), subst(nums, y), subst(nums, z), t, 0, T)
)$
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(%t67)
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